Input system:
Dependent variables:$y_6, y_4, y_1, y_5, y_2, y_3$
Independent variables:a, x_2, b, x_3, x_1
Functions:

$$y_1 = x_3^3\sin(\cos(a\ln(x_2)^2 + bx_3)) + x_3^3 b\ln(x_2)^4 + \sin(a\ln(x_2)^2 + bx_3)^5 + \mathrm{Dv}(g^2, 1, (-1); x_1)$$

$$y_2 = \sin(\cos(a\ln(x_2)^2 + bx_3)) + b\ln(x_2)^4 + \sin(a\ln(x_2)^2 + bx_3)^5 + \mathrm{Dv}(g^2, 1, (-1); x_1)$$

$$y_3 = x_2^2\cos(\sin(a\ln(x_2)^2 + bx_3)) + \cos(a\ln(x_2)^2 + bx_3)^4 + \mathrm{EK}(\sin(\sin(a\ln(x_2)^2 + bx_3)), x_1)^2$$

$$y_4 = x_2^2\cos(\sin(a\ln(x_2)^2 + bx_3)) + \cos(a\ln(x_2)^2 + bx_3)^4 + \mathrm{EK}(\sin(\sin(a\ln(x_2)^2 + bx_3)), x_1)$$

$$y_5 = x_1^2\mathrm{ch}(a\ln(x_2)^2 + bx_3)^2 + x_1^2\sin(a\ln(x_2)^2 + bx_3) + \mathrm{sh}(a\ln(x_2)^2 + bx_3)^5$$

$$y_6 = x_1\mathrm{ch}(a\ln(x_2)^2 + bx_3)^2 + x_1\sin(a\ln(x_2)^2 + bx_3) + \mathrm{sh}(a\ln(x_2)^2 + bx_3)^5$$

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Result:
Dependent variables:$s_6, y_6, s_1, y_4, y_5, y_1, s_{11}, s_{15}, y_2, s_{14}, s_8, y_3, s_{10}, s_{13}, s_{12}, s_3, s_{16}, s_5, s_7, s_9, s_2, s_4$
Independent variables:a, x_2, b, x_3, x_1
Additional variables:

$$s_1 = \ln(x_2)$$

$$s_2 = \mathrm{inv}(x_2)$$

$$s_3 = \cos(as_1^2 + bx_3)$$

$$s_4 = \sin(as_1^2 + bx_3)$$

$$s_5 = \sin(s_3)$$

$$s_6 = \cos(s_3)$$

$$s_7 = \mathrm{Dv}(g^2, 1, (-1); x_1)$$

$$s_8 = \mathrm{DV}(g^2, 1, (-1); x_1)$$

$$s_9 = \cos(s_4)$$

$$s_{10} = \sin(s_4)$$

$$s_{11} = \mathrm{EK}(s_{10}, x_1)$$

$$s_{12} = \mathrm{EKs}(s_{10}, x_1)$$

$$s_{13} = \mathrm{EKc}(s_{10}, x_1)$$

$$s_{14} = \mathrm{EKi}(s_{10}, x_1)$$

$$s_{15} = \mathrm{ch}(as_1^2 + bx_3)$$

$$s_{16} = \mathrm{sh}(as_1^2 + bx_3)$$

Functions:

$$y_1 = x_3^3 bs_1^4 + s_4^5 + x_3^3 s_5 + s_7$$

$$y_2 = bs_1^4 + s_4^5 + s_5 + s_7$$

$$y_3 = s_3^4 + x_2^2 s_9 + s_{11}^2$$

$$y_4 = s_3^4 + x_2^2 s_9 + s_{11}$$

$$y_5 = s_{16}^5 + x_1^2 s_{15}^2 + x_1^2 s_4$$

$$y_6 = s_{16}^5 + x_1 s_{15}^2 + x_1 s_4$$

Equations:

$$\frac{\partial s_1}{\partial a} = 0$$

$$\frac{\partial s_1}{\partial x_2} = s_2$$

$$\frac{\partial s_1}{\partial b} = 0$$

$$\frac{\partial s_1}{\partial x_3} = 0$$

$$\frac{\partial s_1}{\partial x_1} = 0$$

$$\frac{\partial s_2}{\partial a} = 0$$

$$\frac{\partial s_2}{\partial x_2} = (-1)s_2^2$$

$$\frac{\partial s_2}{\partial b} = 0$$

$$\frac{\partial s_2}{\partial x_3} = 0$$

$$\frac{\partial s_2}{\partial x_1} = 0$$

$$\frac{\partial s_3}{\partial a} = (-1)s_4 s_1^2$$

$$\frac{\partial s_3}{\partial x_2} = (-2)s_4 s_1 s_2 a$$

$$\frac{\partial s_3}{\partial b} = (-1)s_4 x_3$$

$$\frac{\partial s_3}{\partial x_3} = (-1)s_4 b$$

$$\frac{\partial s_3}{\partial x_1} = 0$$

$$\frac{\partial s_4}{\partial a} = s_3 s_1^2$$

$$\frac{\partial s_4}{\partial x_2} = 2s_3 s_1 s_2 a$$

$$\frac{\partial s_4}{\partial b} = s_3 x_3$$

$$\frac{\partial s_4}{\partial x_3} = s_3 b$$

$$\frac{\partial s_4}{\partial x_1} = 0$$

$$\frac{\partial s_5}{\partial a} = (-1)s_6 s_4 s_1^2$$

$$\frac{\partial s_5}{\partial x_2} = (-2)s_6 s_4 s_1 s_2 a$$

$$\frac{\partial s_5}{\partial b} = (-1)s_6 s_4 x_3$$

$$\frac{\partial s_5}{\partial x_3} = (-1)s_6 s_4 b$$

$$\frac{\partial s_5}{\partial x_1} = 0$$

$$\frac{\partial s_6}{\partial a} = s_5 s_4 s_1^2$$

$$\frac{\partial s_6}{\partial x_2} = 2s_5 s_4 s_1 s_2 a$$

$$\frac{\partial s_6}{\partial b} = s_5 s_4 x_3$$

$$\frac{\partial s_6}{\partial x_3} = s_5 s_4 b$$

$$\frac{\partial s_6}{\partial x_1} = 0$$

$$\frac{\partial s_7}{\partial a} = 0$$

$$\frac{\partial s_7}{\partial x_2} = 0$$

$$\frac{\partial s_7}{\partial b} = 0$$

$$\frac{\partial s_7}{\partial x_3} = 0$$

$$\frac{\partial s_7}{\partial x_1} = s_8$$

$$\frac{\partial s_8}{\partial a} = 0$$

$$\frac{\partial s_8}{\partial x_2} = 0$$

$$\frac{\partial s_8}{\partial b} = 0$$

$$\frac{\partial s_8}{\partial x_3} = 0$$

$$\frac{\partial s_8}{\partial x_1} = (-1)s_7 p_1 x_1^2 + (-1)s_7 p_2 x_1 + (-1)s_7 p_3$$

$$\frac{\partial s_9}{\partial a} = (-1)s_{10} s_3 s_1^2$$

$$\frac{\partial s_9}{\partial x_2} = (-2)s_{10} s_3 s_1 s_2 a$$

$$\frac{\partial s_9}{\partial b} = (-1)s_{10} s_3 x_3$$

$$\frac{\partial s_9}{\partial x_3} = (-1)s_{10} s_3 b$$

$$\frac{\partial s_9}{\partial x_1} = 0$$

$$\frac{\partial s_{10}}{\partial a} = s_9 s_3 s_1^2$$

$$\frac{\partial s_{10}}{\partial x_2} = 2s_9 s_3 s_1 s_2 a$$

$$\frac{\partial s_{10}}{\partial b} = s_9 s_3 x_3$$

$$\frac{\partial s_{10}}{\partial x_3} = s_9 s_3 b$$

$$\frac{\partial s_{10}}{\partial x_1} = 0$$

$$\frac{\partial s_{11}}{\partial a} = s_{12} s_{14} s_9 s_3 s_1^2$$

$$\frac{\partial s_{11}}{\partial x_2} = 2s_{12}s_{14}s_9s_3s_1s_2a$$

$$\frac{\partial s_{11}}{\partial b} = s_{12}s_{14}s_9s_3x_3$$

$$\frac{\partial s_{11}}{\partial x_3} = s_{12}s_{14}s_9s_3b$$

$$\frac{\partial s_{11}}{\partial x_1} = s_{14}$$

$$\frac{\partial s_{12}}{\partial a} = s_{12}s_{13}s_{14}s_9s_3s_1^2$$

$$\frac{\partial s_{12}}{\partial x_2} = 2s_{12}s_{13}s_{14}s_9s_3s_1s_2a$$

$$\frac{\partial s_{12}}{\partial b} = s_{12}s_{13}s_{14}s_9s_3x_3$$

$$\frac{\partial s_{12}}{\partial x_3} = s_{12}s_{13}s_{14}s_9s_3b$$

$$\frac{\partial s_{12}}{\partial x_1} = s_{13}s_{14}$$

$$\frac{\partial s_{13}}{\partial a} = (-1)s_{12}^2s_{14}s_9s_3s_1^2$$

$$\frac{\partial s_{13}}{\partial x_2} = (-2)s_{12}^2s_{14}s_9s_3s_1s_2a$$

$$\frac{\partial s_{13}}{\partial b} = (-1)s_{12}^2s_{14}s_9s_3x_3$$

$$\frac{\partial s_{13}}{\partial x_3} = (-1)s_{12}^2s_{14}s_9s_3b$$

$$\frac{\partial s_{13}}{\partial x_1} = (-1)s_{12}s_{14}$$

$$\frac{\partial s_{14}}{\partial a} = (-1)s_9s_3s_1^2s_{10}s_{12}^2s_{14}^3 + s_9s_3s_1^2s_{13}s_{14}^2$$

$$\frac{\partial s_{14}}{\partial x_2} = (-2)s_9 s_3 s_1 s_2 a s_{10} s_{12}^2 s_{14}^3 + 2 s_9 s_3 s_1 s_2 a s_{13} s_{14}^2$$

$$\frac{\partial s_{14}}{\partial b} = (-1)s_9 s_3 x_3 s_{10} s_{12}^2 s_{14}^3 + s_9 s_3 x_3 s_{13} s_{14}^2$$

$$\frac{\partial s_{14}}{\partial x_3} = (-1)s_9 s_3 b s_{10} s_{12}^2 s_{14}^3 + s_9 s_3 b s_{13} s_{14}^2$$

$$\frac{\partial s_{14}}{\partial x_1} = (-1)s_{10} s_{14}^3 s_{12}$$

$$\frac{\partial s_{15}}{\partial a} = s_{16} s_1^2$$

$$\frac{\partial s_{15}}{\partial x_2} = 2 s_{16} s_1 s_2 a$$

$$\frac{\partial s_{15}}{\partial b} = s_{16} x_3$$

$$\frac{\partial s_{15}}{\partial x_3} = s_{16} b$$

$$\frac{\partial s_{15}}{\partial x_1} = 0$$

$$\frac{\partial s_{16}}{\partial a} = s_{15} s_1^2$$

$$\frac{\partial s_{16}}{\partial x_2} = 2 s_{15} s_1 s_2 a$$

$$\frac{\partial s_{16}}{\partial b} = s_{15} x_3$$

$$\frac{\partial s_{16}}{\partial x_3} = s_{15} b$$

$$\frac{\partial s_{16}}{\partial x_1} = 0$$